

Storage Connector – Process Report

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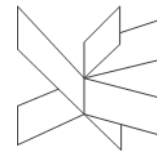
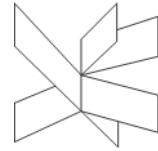


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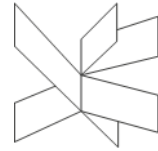
1. Introduction

My motivation for the project was to apply theoretical and practical knowledge about secure distributed systems by designing and implementing a microservice-based, secure, read-only file browser that integrates Google Drive and Microsoft OneDrive into a single unified interface.

The process followed a combination of Unified Process and Kanban to keep the project structured but flexible. Unified Process supports step-by-step, iterative development, while Kanban focuses on continuous flow and adaptability (Kruchten, 2004; Anderson, 2010). Progress was continuously tracked using a task board (see Appendix A), and development advanced iteratively from project initiation to final delivery, following well-established iterative software engineering practices (Sommerville, 2016).

2. Group Work

Since I worked on the project alone, I was responsible for all process disciplines, including Analysis, Design, Implementation, Testing, and Documentation. Working alone meant I did not have to spend time coordinating with others, which helped me keep a clear vision for the project. At the same time, it took a lot of self-discipline, careful planning, and regular self-checks. Since I could not discuss ideas with a team, I made sure to think through my decisions and checked important choices with my supervisor when necessary.



3. Project initiation

3.1 Problem Domain and Scope

At the start of the project, I identified the main problem: users lose productivity when managing files across different cloud storage platforms. From this, I created a problem statement focused on providing unified file access while preserving user security and privacy.

To make sure the project could be completed within the bachelor's timeframe, I set clear boundaries for its scope:

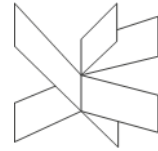
- I chose to integrate only Google Drive and Microsoft OneDrive.
- The system allows users to access files in read-only mode.
- Advanced features like editing, sharing, were not included.

These boundaries helped keep the project manageable and allowed me to focus on the main learning goals.

3.2 Planning and Risk Awareness

Next, I created a high-level schedule and estimated about 300 hours of work. I looked at risks such as OAuth complexity, API limits, and time management, and planned to handle them by adding buffer time and using incremental integration.

I also thought about ethical issues, making sure to protect user privacy, handle tokens securely, and limit data access as much as possible by design.



4. Project Execution

4.1 Applied Methods

To manage the project, I decided to use both the Unified Process and Kanban. Unified Process gave me structure with its phases: Inception, Elaboration, Construction, and Transition, while Kanban helped me organize daily tasks using a board with Not Started, In Progress, and Done columns. This setup let me keep improving requirements and design while steadily moving the project forward.

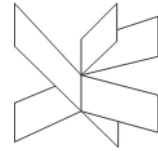
4.2 Iterative Development

Development progressed incrementally:

1. Establishing project structure and authentication foundations.
2. Implementing OAuth 2.0 with PKCE for each provider.
3. Designing a unified file model.
4. Aggregating files across providers.
5. Implementing frontend workflows.

Testing and documentation were performed continuously rather than as final phases. When issues arose, I revisited analysis or design decisions, demonstrating an iterative and reflective process.

Since I managed every part of the project, I kept a clear link between requirements, use cases, implementation, and testing. Seeing the whole process helped me make better decisions about architecture and security.



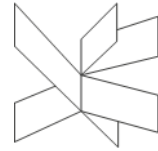
5. Personal Reflections

This project helped me get much better at designing secure distributed systems, especially with OAuth 2.0, identity management, and microservice architectures. Since I worked alone, I had to plan, carry out, and review every part of the project myself. This experience taught me how to organize my work, set priorities, and make technical decisions on my own.

One key thing I learned was how much early design choices affect later work, especially when it comes to implementation and security. Making sure third-party services were integrated safely turned out to be harder than I expected. This showed me how important it is to really understand outside systems and what they can and cannot do.

If I did a similar project again, I would spend more time on security and load testing to see how the system works in real situations. I would also start usability testing earlier to check my ideas about how users interact with the system before things are too set in place.

This project showed me that I am good at thinking about systems as a whole and keeping track of complex setups. It also made me realize that getting feedback and checking my work earlier could have made things go more smoothly. Working alone helped me become more independent, but it also made me value teamwork. When people work together and share ideas, decisions are often quicker and more complete, which I missed while working by myself.



6. Reflect on Supervision

Supervision was helpful and encouraging throughout the project. I reached out to my supervisor mainly when I was unsure about architectural choices, security issues, or the limits of the project's scope.

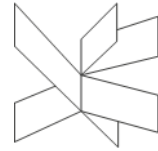
Instead of meeting often, I used supervision when I needed to check my assumptions, confirm my technical approach, or get advice on different options.

In future projects, I would aim to prepare even more concrete questions before supervision sessions to maximize their value.

7. Conclusion

This project showed me that having a process that is both organized and adaptable is important for working alone. From my experience, I recommend keeping track of your tasks, regularly thinking about your decisions, and checking complex choices early on.

Using both the Unified Process and Kanban helped me handle a development-focused project on my own and meet my academic and technical goals.



8. References

Appendix A – Product Backlog

Kruchten, P. (2004). The Rational Unified Process: An Introduction. Addison-Wesley.

Anderson, D. J. (2010). Kanban: Successful Evolutionary Change for Your Technology Business. Blue Hole Press.

Sommerville, I. (2016). Software Engineering (10th ed.). Pearson.